

Arbitrage and Convergence: Evidence from Mexican ADRs

by

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ABSTRACT

This paper investigates the convergence between the prices of ADRs and the prices of the Mexican traded shares using a sample of 21 dually listed shares (listed in Mexico and in the U.S.). Since both markets have similar trading hours in Mexico, standard arbitrage considerations should make persistent deviation from price parity rare. We estimate the convergence of the ADRs and the locally traded shares using non-linear models. We reject the linear adjustment model and, based on different tests, we select the ESTAR model. We find that price spreads tend to die out quickly; for 15 out of 21 pairs, it takes about 1 day for the ADR-underlying price spread to be reduced by half. Four firms, however, have high half-life estimates (seven days or more), and in general correspond to companies that display very low average daily volume, and thus, arbitrage might be difficult to execute. The results of Spearman correlation tests confirm this finding as most liquidity market indicators are positively correlated to the speed of convergence to parity. The average half-life is 3.1 business days. By allowing non-linear adjustments, the average half-life is reduced by more than 57% when compared to the standard linear arbitrage model.

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I. Introduction

The market for American Depositary Receipts (ADRs) has developed for various reasons most of which are analyzed in the literature, including value maximization, portfolio diversification, investor recognition, and overcoming market segmentation. ADRs are negotiable certificates traded in the U.S. financial markets; they simply represent the shares of foreign country firms. U.S. commercial banks serve as the depository units for the ADRs. Thus, although trading ADRs in the U.S. is U.S. dollar denominated, it should be equivalent to trading the foreign firms' shares without actually trading them in their respective local markets. Price and return reaction to cross market listing, possible arbitrage opportunities, and the difference between ADR returns and the returns on their local counterpart shares are some of the issues raised by many researchers. For excellent reviews of this growing body of literature, see Karolyi (1998, 2004).

In this paper, we focus on the possible arbitrage opportunities that the ADR market provides. In the absence of direct or indirect trading barriers, there should not be significant differences between the return distribution of locally traded shares and that of the U.S. traded ADRs. That is, ADRs and their underlying shares are expected to be perfect substitutes and no arbitrage opportunities should prevail. Many papers deal with the issue of international barriers to trading, investments, and cash flows movements. Stulz (1981) develops a model of investment with international trading barriers. Eun and Janakiramanan (1986) describe many of the barriers that existed at that time. More recently, Stulz and Wasserfallen (1995) analyze a case of market segmentation in

Switzerland, and Domowitz, Glen and Madhavan (1997) develop a model of market segmentation based on cash flows restrictions.

There is a good body of literature that studies the potential arbitrage opportunities that the ADRs create. If prices between the ADRs and their underlying shares differ substantially, arbitrage opportunities will arise. The early studies by Maldonado and Saunders (1983), Kato, Linn and Schallheim (1991), Park and Tavakkol (1994), Miller and Morey (1996) and Karolyi and Stulz (1996) concluded that ADRs do not present any arbitrage opportunities. The only early study that did find some arbitrage opportunities is by Wahab, Lashgari and Cohn (1992). Substantial deviations from arbitrage pricing are consistent with other studies in the literature of dually-listed shares, such as Rosenthal and Young (1990), and Froot and Dabora (2003). More recent work by Hong and Susmel (2003) and De Jong, Rosenthal and van Dijk (2004) show that deviation from theoretical arbitrage-free prices between ADRs and their underlying shares can be substantial. Furthermore, Gagnon and Karolyi (2003) quantify sizable price deviations from arbitrage-free pricing. These significant deviations from arbitrage pricing have been attributed to market inefficiencies, see Mullainathan and Thaler (2000) and Barberis and Thaler (2003).

In this paper, we study the price convergence between the ADRs and their underlying shares in a nonlinear framework. We focus on Mexican ADRs because the trading hours in Mexico and New York are almost identical, thus, the results should not be affected by possible lead lag informational impact, as analyzed by Kim, Szakmary and Mathur (2000).¹ Numerous studies in this area have, implicitly, focused on linear

¹ Kim, Szakmary and Mathur (2000) show that uncertainty induced by non-synchronous trading of the ADRs and the underlying securities, impede arbitrage activity.

convergence to arbitrage parity.² Given the complexity of rules and indirect transaction costs, however, non-linear adjustments to price parity deviation are more likely to occur. Thus, in this paper, we use two popular non-linear models for our adjustment specification: the exponential smooth transition autoregressive (ESTAR) and the logarithmic smooth transition autoregressive (LSTAR). From our estimation results, first, we reject the linear adjustment model; and, second, based on different tests, we select the ESTAR model. Using the ESTAR model, we are able to estimate the half-life of different shocks. We find that price spreads tend to die out quickly, for 15 out of 21 pairs, it takes about 1 day for the ADR-underlying price spread to be reduced by half. Four firms, however, have high half-life estimates (seven days or more). Three of these four firms correspond to companies that display very low volume, and thus, arbitrage might be difficult to execute. The average half-life is 3.1 business days. By allowing non-linear adjustments, the average half-life is reduced by more than 57%, when compared to the standard linear model.

This paper is organized as follows. Section II presents a brief literature review. Section III presents a simple arbitrage model and different non-linear adjustment models. Section V estimates the non-linear model and analyzes the conversion path to arbitrage parity. Section VI concludes the paper.

II. Literature Review

There is a growing body of literature that studies the potential arbitrage opportunities that cross-listed shares create. If prices between the local shares and their

² See for instance Kim et al. (2000) for VAR and SUR approaches to analyze the speed of adjustment of ADR prices; and Gagnon and Karolyi (2003) for a standard AR model.

cross-listed shares differ substantially, arbitrage opportunities will arise. As pointed out in the introduction, the early studies conclude that arbitrage opportunities are non-existent for cross-listed shares and thus cross-listed shares are priced according to arbitrage parity. The only early study that did find some arbitrage opportunities is by Wahab, Lashgari and Cohn (1992). Some recent works, however, have found a significant deviation from arbitrage price parity. Froot and Dabora (1999), studying the pricing of two dual-listed companies, Royal Dutch and Shell, and Unilever N.V. and Unilever PLC, find a large and significant price deviation from arbitrage parity. Gagnon and Karolyi (2003) quantify sizable price deviations from arbitrage-free pricing between ADRs and their underlying assets. Gagnon and Karolyi (2003) document the existence of large price deviations for many of the 581 ADR-underlying pairs they study. They estimate discounts of up to 87% and premia of up to 66%. Gagnon and Karolyi (2003), after taking into account direct and indirect transaction costs still find the price deviations to be exceeding reasonable measures of transaction costs. Still, they mention that the complexity of rules in the ADR-underlying arbitrage precludes definite conclusions about potential market inefficiencies.

Large price deviations from arbitrage-price parity, do not necessarily imply arbitrage profits are possible. Transaction costs, capital control restrictions, conversion rules, and lack of liquidity might make arbitrage very difficult. De Jong et al.(2003) and Hong and Susmel (2003) attempt to construct realistic arbitrage strategies to see whether arbitrage is possible. De Jong et al.(2003) extend the sample to 13 dual-listed companies and show that for every individual dual-listed company, deviations from arbitrage price parity are large. They design investment strategies for exploiting these mispricings. They

find that some arbitrage strategies in all dual-listed companies produce excess returns of up to 10% per annum on a risk-adjusted basis, after transaction costs and margin requirements. Interestingly, they document that for some dual-listed companies it takes 9 years for prices to converge to arbitrage price parity. Hong and Susmel (2003) study simple arbitrage profits for ADR-underlying pairs. They find that pairs-trading strategies deliver significant profits. The results are robust to different profit measures and different holding periods. For example, for a conservative investor willing to wait for a one-year period, before closing the portfolio pairs-trading positions, pairs-trading delivers annualized profits over 33%.

In the related line of research lays price discovery literature. This literature deals with cointegrated pairs in a standard linear adjustment framework. For example, Eun and Sabherwal (2003) apply a standard error correction model to study price discovery shares for 62 Canadian shares cross-listed in the NYSE. They find a significant price deviation from arbitrage parity. They find that the price adjustments of U.S. prices to deviation from Canadian prices are significantly larger in absolute value. They also find that trading volume in the U.S. is the most important variable in the determination of relative information contribution of the two markets. Using intradaily data and a similar methodology, but for only three German firms, Gramming, Melvin, and Schlag (2001) find that the majority of the price discovery is done at home (Germany), but following a shock to the exchange rate, almost all of the adjustment comes through the New York price. A related model, but using non-linear adjustment dynamics, is estimated by Rabinovitch et al. (2003). Using a non-linear threshold model for 20 Chilean and Argentine cross-listed stocks, Rabinovitch et al. (2003) estimate transactions costs and

show that transaction costs play an important role in the convergence of prices of ADRs and their underlying securities. Thus, using the non-linear model, they are able to estimate different arbitrage adjustment processes for Chile and Argentina. They find that capital control measures and liquidity significantly affect the price adjustment process, through increasing transactions costs. Melvin (2003) and Auguste et al. (2004) also find that capital movement restrictions can seriously affect the arbitrage price parity, especially during economic and currency crisis.

III. Non-linear convergence and Arbitrage Models

As mentioned in the introduction, the existence of capital controls and institutional features, such as a lack of short-selling, will have an economic impact on arbitrage opportunities. It is quite possible that traders design contracts and find loopholes, which enable them to avoid the effect of capital restrictions. One indirect way to check the effect of these restrictions is to measure arbitrage opportunities. If there are significant arbitrage opportunities, then the capital restrictions have economic implications. This is the path chosen by Hong and Susmel (2003) and De Jong et al. (2003). These papers design simple arbitrage strategies to study the existence of significant profits. Another indirect way to analyze the effect of the above mentioned restrictions on arbitrage opportunities is to study the convergence path of prices to arbitrage parity. Traditionally, the dynamic structure of the price disparity between the ADRs and their underlying shares has been mostly analyzed in a linear framework. This linear framework is counterintuitive since, once arbitrage is triggered, arbitrage

opportunities may disappear very slowly and always at the same speed.³ One way to address this issue is to consider that, under certain conditions, price differences should converge rather faster to parity. This can happen when the convergence dynamics are governed by a nonlinear process. In this paper, we start by assuming that small deviations from arbitrage-free prices between ADRs and their underlying shares may be considered negligible to generate arbitrage activities, notably when transactions and other related trading costs are not covered by the price spread. In this case, the price spread would behave as a near unit root process and would not converge to parity in a linear framework. Rabinovitch et al. (2003) use a threshold model to study the convergence of the returns of the ADR-underlying pair. In this paper, we assume that arbitrage forces will likely increase in pressure as the ADR-underlying pair moves further away from arbitrage parity, or long run equilibrium. That is, we assume that the dynamics of arbitrage in international markets should be analyzed in a nonlinear framework.⁴

Let $P_t(A)$ represent the price of ADRs and $P_t(L)$ the price of underlying (locally) traded shares at time t . The relationship between both prices, under the arbitrage-free condition, with absence of transaction costs is specified as:

$$P_t(A) = b S_t P_t(L) \quad (1)$$

where S_t denotes the nominal exchange rate at time t , and b the bundling price ratio.

³ See Eun and Sabherwal (2003)

⁴ See for instance Dumas (1992), Uppal (1993), Sercu et al. (1995), Coleman (1995), Obstfeld and Taylor (1997). These articles find that market frictions create an inactive transaction band within which small deviations from purchasing power parity prevent the real exchange rate to mean revert. The non-reversion is attributed to excess of the marginal costs of market arbitrage vis-à-vis its marginal benefits. However, arbitrage opportunities exist only for large deviations outside the inactive band. In deed, traders have tendency to postpone entering the market until enormous arbitrage opportunities open up.

Let κ measure the transaction costs faced by arbitrageurs. Arbitrage activities will arise under one of the following conditions:

$$\frac{bS_t P_t(L)}{P_t(A)} < 1 - \kappa \quad (2)$$

$$\frac{bS_t P_t(L)}{P_t(A)} > 1 + \kappa \quad (3)$$

Provided that κ is small, inequalities (2) and (3) can be expressed in the following logarithmic form:

$$|q_t| > \kappa \quad (4)$$

where $q_t \equiv p_t^L - p_t^A + s_t + b$ represent the price spread between the locally traded shares and ADRs. Suppose that arbitrageurs believe in a long-run arbitrage-free equilibrium between the local shares and the ADRs; that is, equality (1) holds true, and assume that on a given day the local price is larger than the ADR price by more than the transaction costs associated with trading, i.e., $q_t > \kappa$. Arbitrageurs will subsequently have an incentive to invest in the ADR market and, arbitrage activities will then create a reversion to the long-run equilibrium spread. The focus of this paper is on the time series behavior of q_t .

III.1 Modeling Nonlinear Adjustment of Price Spread

One parametric model suggested in the theoretical literature to capture this type of nonlinear adjustment is the threshold autoregressive (TAR) model pioneered by Tong (1990) in which regime changes occur abruptly. In this case, beyond a threshold level of the absolute deviation from price parity, price spread converges to arbitrage parity while

displaying a unit root behavior elsewhere.⁵ A problem with this approach is that it has two very distinct regimes: outside the threshold (where arbitrage happens) and inside the threshold (where there is no arbitrage). The change from one regime to the other is abrupt and it presumes the same speed of adjustment outside the threshold. This assumption is probably unrealistic, since large deviations outside the threshold will probably trigger more arbitrageurs to act than small deviations outside the threshold.⁶ An alternative way of tracking this nonlinear adjustment process is through the smooth transition autoregressive (STAR) model instigated by Granger and Teräsvirta (1993) and Teräsvirta (1994) in which transitions between regimes are assumed to occur gradually. In the empirical nonlinear literature, the Exponential STAR (ESTAR) and the Logistic STAR (LSTAR) are two popular models used respectively for symmetric and asymmetric adjustments. The adjustment structure of both models depends on the magnitude of the departure of the underlying process from its equilibrium. In this paper, we use STAR models, as some time should elapse before arbitrageurs take an appropriate position. Such arbitrage opportunities are necessarily lagged over a time period.⁷ A STAR model of order p for the univariate time series q_t can be formulated as:

$$q_t = \Psi_1' x_t + \Psi_2' x_t \Phi(z_t; l, m) + e_t, \quad \lambda > 0 \quad (5)$$

or

$$q_t = \mu + \sum_{j=1}^p \Psi_{1j} L^j (q_t - \mu) + \left[\sum_{j=1}^p \Psi_{2j} L^j (q_t - \mu) \right] [\Phi(z_t; l, m)] + e_t, \quad \lambda > 0 \quad (5')$$

⁵ Using the TAR models, Obstfeld and Taylor (1997), and O'Connell and Wei (1997) reported evidence of nonlinear price adjustment induced by the presence of transaction costs.

⁶ In addition, U.S. market makers, who provide liquidity in the NYSE, have an obligation to provide price continuity. Thus, U.S. market makers provide smooth transitions in stock prices.

⁷ Since the price of ADRs and the price of their underlying shares have potentially unit roots and hence are cointegrated, a nonlinear error correction type of model could be estimated, see for instance Dwyer et

where the sequence $\{q_t\}$ is a stationary and ergodic process and the error term, e_t , follows an identical and independent distribution, with zero mean and constant variance s^2 . The independent variable x_t is defined as $x_t = (1, \tilde{x}_t)'$ with $\tilde{x}_t = (q_{t-1}, q_{t-2}, \dots, q_{t-p})'$ and $\Psi_i = (\Psi_{i0}, \Psi_{i1}, \dots, \Psi_{ip})$, $i=1,2$, denotes the autoregressive parameters vector of dimension p of an AR(p); L is a lag operator; the transition function $\Phi(z_t; l, m)$ determines the degree of convergence. A transition function for symmetric adjustments that possesses appealing mathematical properties has the general exponential specification⁸:

$$\Phi(z_t; \lambda, \mu) = 1 - \exp\left\{-l(z_t - \mu)^2 / \hat{s}_{z_t}^2\right\}, \quad \lambda > 0 \quad (6)$$

where, z_t , the transition variable is assumed to be a lagged endogenous variable $z_t = q_{t-d}$ (or a linear combination of lagged endogenous variables, $z_t = \alpha' \tilde{q}_t$) for which d is the delay lag, a certain nonzero integer ($d > 0$), that determines the lagged time between a shock and the response by the process. While λ determines the speed of transition between regimes, with lower values implying slower transition, μ can be interpreted as the arbitrage parity, equilibrium level.

The transition function $\Phi(\cdot)$ is continuously differentiable with respect to the scale parameter λ and bounded between 0 and 1, $\Phi: \Re \times \Re^+ \times \Re \rightarrow [0,1]$ with the properties $\Phi(0)=0$ and $\lim_{|q_{t-d}| \rightarrow \infty} \Phi(\cdot) = 1$. F is then symmetrical around the equilibrium level (mean).

Substituting (6) into (5), the ESTAR model can be written as:

$$q_t = \mu + \sum_{j=1}^p \Psi_{1j} L^j (q_t - \mu) + \left[\sum_{j=1}^p \Psi_{2j} L^j (q_t - \mu) \right] [1 - \exp\left\{-l(z_t - \mu)^2 / \hat{s}_{z_t}^2\right\}] + e_t \quad (7)$$

al.(1996), Martens et al. (1998), and Taylor et al. (2000).

⁸ The sample variance of the transition variable is used to scale the argument of the exponential as suggested by Granger and Teräsvirta (1993, p.124). The scaling indeed enables a stability improvement of the nonlinear least

$$\text{or} \quad q_t = \Psi_1' x_t + \Psi_2' x_t [1 - \exp\{-\lambda (z_t - \mu)^2 / \hat{s}_{z_t}^2\}] + e_t \quad (7')$$

Since the transition function is bounded between zero and one, the inner regime is characterized by $q_{t-d} = \mu$, when $\Phi(.) = 0$. The ESTAR model (4) then degenerates to a standard linear AR (p):

$$q_t = \mu + \sum_{j=1}^p \Psi_j L^j (q_t - \mu) + e_t, \quad (8)$$

The outer regime is characterized by an extreme deviation from the equilibrium,

$\lim_{|q_{t-d}| \rightarrow \infty} \Phi(.) = 1$, in which case model (4) converts to a different AR(p) representation:

$$q_t = \mu + \sum_{j=1}^p (\Psi_{1j} + \Psi_{2j}) L^j (q_t - \mu) + e_t, \quad (9)$$

The model displays global stability provided $\sum_{j=1}^p (\Psi_{1j} + \Psi_{2j}) < 1$, although it is possible

that $\sum_{j=1}^p \Psi_j \geq 1$ implying then q_t may follow a unit root process or even explodes around

the arbitrage free parity level.

The LSTAR model uses the logistic function, instead of an exponential function, to model the transition function $\Phi(.)$. Thus, after substituting in (5'), the LSTAR model can be written as:

$$q_t = \Psi_1' x_t + \Psi_2' x_t [1 / (1 + \exp\{-\lambda (z_t - \mu) / \hat{s}_{z_t}\})] + e_t, \quad \lambda > 0.$$

III.2 Testing and Estimation of STAR Models

squares estimation algorithm, a fast convergence, and an interpretation and comparison of ? estimates across equations in a scale-free environment.

The starting point in modeling a STAR specification consists of an adequate choice of the autoregressive parameter p , the delay parameter, d , and the conduct of a sequence of tests of linearity along with other diagnostic tests⁹. If the null hypothesis of linearity (AR model) is rejected, the model can be usefully specified as a nonlinear alternative ESTAR or LSTAR.¹⁰

The choice of the autoregressive parameter, p , is based on the Akaike information criterion (AIC) and, also, by inspection of the partial autocorrelation function (PACF)¹¹ using a 95% confidence interval band. In order to specify the delay parameter, d , a sequence of linearity tests is carried out for different ranges of d with $1 = d = D$ considered appropriate. If the null hypothesis of linearity is rejected at a pre-specified level for more than one value of d , then d is determined at $d = d^*$ such that: $d^* = \text{Arg}\{\text{Min } p(d)\}$ for $1 = d = D$, where $p(d)$ denotes the (p-value) of the selected test. The correct choice of d is important for the test to have a maximum power. For this paper, we set the maximum value of d equal to 5 business days as it seems unreasonable to argue that it would take more than 5 days for the price spread to start adjusting if there is an arbitrage activity.

We test for the presence of nonlinearity in the price spread between the local assets and their corresponding ADRs using the Lagrange Multiplier (LM) tests proposed by Luukkonen et al.(1988) (hereafter, the LSP procedure); Granger and Teräsvirta (1993) and Teräsvirta (1994) (hereafter, the TP procedure); and Escribano and Jordá (1999) (hereafter, the EJP procedure). For each test, we conduct a heteroskedasticity-consistent

⁹ See Van dijk et al.(2002) for a survey of the different modeling procedures for STAR models.

¹⁰ For instance see Luukkonen et al. (1988a) for testing linearity against STAR models and the related problems.

¹¹ The AIC tends to under-parameterize an AR model. Therefore, a choice of the lag length based solely on AIC is more likely to bias linearity tests in that the presence of remaining serial correlation reduces the test power.

specification since neglecting heteroskedasticity can seriously affect the power of LM tests, by rejecting too often the null hypothesis. We use the specification tests developed by Wooldridge (1990, 1991) that do not require the specification of the form of the heteroskedasticity¹².

Using equation (5'), we can test the null hypothesis of linearity, by testing $H_0: \Psi_2 = 0$ against the alternative hypothesis $H_1: \Psi_{2,j} \neq 0$ for at least one $j \in \{0, \dots, p\}$. However, under the null, the transition function's parameters λ and μ are unidentified. It is suggested in the literature to use a Taylor series expansion (third order) of the transition function $\Phi(q_t; \lambda, \mu)$ around zero to overcome the non-identification issues.¹³ The re-parameterization of equation (5') yields the following artificial regression:

$$q_t = \beta_{00} + \sum_{j=1}^p \beta_{0j} q_{t-j} + \sum_{j=1}^p \beta_{1j} q_{t-j} q_{t-d} + \sum_{j=1}^p \beta_{2j} q_{t-j} q_{t-d}^2 + \sum_{j=1}^p \beta_{3j} q_{t-j} q_{t-d}^3 + v_t \quad (10)$$

where $\beta_j = (\beta_{0j}, \beta_{1j}, \beta_{2j}, \dots, \beta_{pj})$ with $j=1,2,3$ are function of the AR coefficients vector $(\Psi_{i0}, \Psi_{i1}, \dots, \Psi_{ip})$, $i=1,2$ and the transition function parameters λ and μ . Thus, assuming d is known, the null hypothesis of the linearity test can be written as $H_0: [\beta_{1j} = \beta_{2j} = \beta_{3j} = 0]$, with $j=\{1,2,\dots,p\}$. For large samples, the derived test statistic denoted NLM3 follows a χ^2 distribution with $(p+1)$ degrees of freedom. We also use the non-linearity tests developed by Escribano and Jordá (1999) that account for the fourth power of the transition variable¹⁴. The underlying auxiliary regression is:

¹² Van Dijk et al. (1999) showed that in the presence of additive outliers, LM tests for STAR nonlinearity tend to incorrectly reject the null hypothesis of linearity; they then developed outliers-robust tests. We used such tests along with the heteroskedasticity tests, but there were no major changes for our sample.

¹³ See Saikkonen and Luukkonen (1988a) and Teräsvirta (1994).

¹⁴ The argument, indeed, is that when the variance of the error terms is large, causing the data to be asymmetrically distributed around the equilibrium value, LSTAR (a nonlinear model) will be wrongly detected by the test more frequently.

$$q_t = \beta_{00} + \sum_{j=1}^p \beta_{0j} q_{t-j} + \sum_{j=1}^p \beta_{1j} q_{t-j} q_{t-d} + \sum_{j=1}^p \beta_{2j} q_{t-j} q_{t-d}^2 + \sum_{j=1}^p \beta_{3j} q_{t-j} q_{t-d}^3 + \sum_{j=1}^p \beta_{4j} q_{t-j} q_{t-d}^4 \quad (11)$$

The null hypothesis of linearity is then: $H_0: \beta_{1j} = \beta_{2j} = \beta_{3j} = \beta_{4j} = 0$, with $j=1, \dots, p$. The resulting test statistic, denoted NLM ,⁴ follows a chi-squared distribution with $4(p+1)$ degrees of freedom for large samples. The rejection of the null hypothesis will indicate the presence of nonlinearity.

III.3 Model Selection: Testing ESTAR Against LSTAR

Once a nonlinear specification is found adequate, the next task is to choose between the ESTAR and the LSTAR models. Teräsvirta (1994) suggests the use of the artificial regression (10) to perform a LM test of the ESTAR specification against the alternative of the LSTAR specification. In fact, the significance of cubic terms in equation (10) will not indicate the ESTAR adjustment in that the third order Taylor expansion of the transition function of an ESTAR model has a quadratic form (U-shape). The cubic terms will rather signal a LSTAR type of adjustment (asymmetry). In other words, if the null hypothesis $H_{02}: \beta_{2j} = 0 \mid \beta_{3j} = 0$ with $j=1, \dots, p$ is rejected, ESTAR model is selected based on the smallest p-value, whereas the rejection of the null hypothesis $H_{02}: \beta_{3j} = 0$ with $j=1, \dots, p$ leads to the selection of the LSTAR model. Escribano and Jordá (1999) also develop a LM-type test to discriminate between LSTAR and ESTAR using the artificial Equation (11) and conditional on prior rejection of linearity. The selection procedure is as follow: Let LM^{EST} denote the F-test of the null hypothesis $H_{0E}: [\beta_{2j} = \beta_{4j} = 0]$ with $j=1, \dots, p$ for ESTAR, and LM^{LST} the null hypothesis

$H_{0L}: [\beta_{1j} = \beta_{3j} = 0]$ with $j=1, \dots, p$ for LSTAR. The relative strength of the rejection of each hypothesis is then compared. If the minimum p-value corresponds to LM^{LST} , LSTAR is selected, if it rather corresponds to LM^{EST} , the select model is ESTAR.

IV. The Data

The data analyzed in this paper are the daily prices on twenty one locally traded firms from Mexico, obtained from Datastream. To be part of our sample, the ADR has to be Level III or Level II. The sample periods are different for the different firms, depending on the dates for which ADRs started trading on these firms on the U.S. market.

Table 1 presents the twenty one firms and the sample period for each of them. Table 2 exhibits several statistics for each firm: Market Capitalization (MC), average daily volume since inception (Volume), the number of freely traded shares in the hands of the public (Float), and the short-ratio, which is calculated as the short interest for the current month divided by the average daily volume. In the last four columns of Table 2, we also present summary statistics for the theoretical price parity (in %):

$$QPar_t = (b S_t P_t(L) / P_t(A) - 1) * 100.$$

Analyzing the statistics for $QPar_t$, we observe evidence for autocorrelation. We also tend to observe a negative relation between liquidity and departure from theoretical price parity: the less liquid a stock is, the bigger the departures from price parity, as shown by the mean and maximum and minimum statistics.

V. Results

The lag selection is based on both the AIC and the partial autocorrelation

functions (PACF). Figure 1 displays the PACF for selected firms with a 95% confidence interval band.¹⁵ It indicates that for most series, only the first, second or rarely third autocorrelation coefficients are significant at the 5% level. Therefore, the maximum AR used is 3, which seems to purge the residuals series from serial correlation. Although a more parsimonious model would be preferable, we also tried with $p > 5$, with $d = \{1, 2, \dots, 10\}$, to check for plausible high order AR and for the presence of serial correlation in q_t ; the results are very similar to the ones presented below.

Table 3 reports p-values for the standard and heteroskedasticity-consistent test statistics NLM3 and NLM4 for linearity tests, and the test statistics NLM2, LM^{LST} and LM^{EST} for choosing between ESTAR and LSTAR. Panel A of Table 3 shows all the test results for one firm, TMX.¹⁶ Panel B of Table 3 shows a summary of the test results for all the other firms. The first column of Table 3 displays the transition variable the choice of which includes the lagged endogenous variables $z_t = q_{t-d}$ with different values of the delay parameter $d = \{1, 2, \dots, 5\}$ ¹⁷.

Using the results on the first panel of Table 3 for TMX, we select $d=2$, as the results indicate the smallest p-values (corresponding to NLM3 and NLM4) for both tests; this implies the strongest rejection of the AR linear hypothesis. Also for $d=2$, the ESTAR model is selected over the LSTAR model since the p-value of the LM^{EST} test is smaller

¹⁵ The bands represent $\pm 1.96/\sqrt{T}$, with T being the sample size.

¹⁶ The results for the other firms can be reported similarly, but are not included to save space. They are available under request.

¹⁷ The tests are performed with values of the delay parameter, $d = \{1, 2, \dots, 10\}$, yet we report the tests statistics for $d = \{1, 2, \dots, 5\}$ since $d = \{6, \dots, 10\}$ do not alter the choice of d and are less relevant for the convergence of a daily price spread series. We also used as the transition variable, z_t , the first lag of the average absolute volatility, $v_{t,k}$ as suggested by LeBaron (1992) as: $v_{t,k} = \frac{1}{k} \sum_{i=0}^{k-1} |q_{t-i}|$, where k is the number of days, with a maximum of 5 business days. The tests selected $v_{t,k}$ as adequate transition variables for six stocks.

than the p-value of the LM^{LST} test, for both versions of the test. Note that the p-value of the LNM2 test confirms this selection. We follow this process for the other firms. Based on the standard LM test statistics NLM3 and NLM4, reported in the second panel of Table 3 for all the firms, the null hypothesis of linearity can be rejected for any values of d and corresponding transition variables, at the 1% level. When we allow for the heteroskedasticity-consistent robust tests, the null hypothesis of linearity is rejected only for some transition variables and values of d ($d=1,2$). For example, for FMX the results of the heteroskedasticity-robust test indicate that the transition variables $z_t = q_{t-1}, q_{t-2}$, and $v_{t-1,4}$ are adequate transition variables as the corresponding p-values are smaller. The results from the test statistic NLM4, computed using the Escibano and Jordá test, generally confirm this selection. Finally, the p-values of the LM statistics (standard or heteroskedasticity-consistent) NLM2 suggest an ESTAR model is the more appropriate model. Comparing relative strength of the tests LM^{EST} and LM^{LST} , the minimum p-values correspond to LM^{EST} , indicating a choice in favor of the ESTAR model. In most cases, the LM^{EST} is significant at the 1% level for $d=2$. Thus, based on the decision rules of the Teräsvirta procedure and the Escibano and Jordá procedure, the ESTAR model with a delay, $d=2$ should be an adequate model specification for FMX return spread. Carrying on identical analyses for the other firms, we report the results of linearity tests and model selection in Table 3.

V.I Nonlinear Estimation Results

An ESTAR (p) specification is selected for all firms with $p=\{1, 2, 3\}$ and the resulting models are estimated by nonlinear least squares¹⁸. The following restrictions consistent with the application of ESTAR, $\Psi_j = -\Psi_j^*$ ($j=1,...,p$), $\Psi_{1j} + \Psi_{2j} = 1$ (for those models with $p>1$), and $\mu = 0$ are also tested using the likelihood ratio tests. If those restrictions cannot be rejected, then they are imposed, and the model is re-estimated; in such cases the final model is governed by λ , the speed of transition between regimes.¹⁹ The model estimates, the likelihood ratio, and residuals diagnostic statistics are presented in Table 4. In column ten, we report the p-value associated with the likelihood ratio statistic, LR(k). The LR(k) statistics indicate the restrictions cannot be rejected at the standard 5% level for all series, while the number of restrictions imposed varies from one firm to another. For example for the firm AMX, the p-value of LR(4) is 0.561, so that we failed to reject four restrictions²⁰, and the model was re-estimated by imposing these restrictions. In fourth column of Table 4, we report the estimates of the transition parameter, λ . With one exception, the estimates of λ are all significant based on the individual t-ratio (the standard deviation is in parenthesis).²¹ The size of λ changes from 2.971 to 0.315. It is worthy noticing that firms with a higher estimate of λ tend to have higher average daily volume and market capitalization. Whereas firms for which the price spread series exhibits a lower speed of adjustment coefficients, such as ICM ($\lambda=0.317$),

¹⁸ See Gallant and White (1988) for further discussions.

¹⁹ These restrictions imply the presence of a unit root behavior of price spread around the parity level with a potential to revert to parity.

²⁰ $\mu = 0$, $\Psi_{11} + \Psi_{12} = 1$, $\Psi_{21} = -\Psi_{11}$, and $\Psi_{22} = -\Psi_{12}$

²¹ However, Taylor et al. (2001) point out that the significance of λ estimate based on individual t-ratios should be checked for robustness. Indeed, technical problems emerge under the null hypothesis that $\lambda = 0$, since "each series follows a unit root process, which complicates the test procedure similarly to the way in which the distribution of a DF statistic cannot be assumed to be student's t."

GMK ($\lambda=0.361$), TMM ($\lambda=0.315$), CEL ($\lambda=0.496$), and SIM ($\lambda=0.514$), tend to have lower average daily volume and market capitalization. All estimated values of λ support the ESTAR specification and provide evidence of nonlinear convergence of price spread series to parity.

We also conducted specification tests for our ESTAR model. The residuals diagnostic statistics for the estimated equations are reported in the last two columns of Table 4. LMNA AR(1-6) is a LM-test statistics for testing the null hypothesis of no serial correlation in the residuals of order 1, up to 6 and NL_{Max} denotes the maximum LM-test statistic of no additive nonlinearity with the delay length in the range from 3 to 6.²² The associated p-values indicate that we cannot reject those null hypotheses for all firms at the 5% level or better. Therefore, an ESTAR specification seems adequate for the price spread series.

V.II Estimated Transition Functions

The transition function measures the magnitude of deviations of the price spread from its arbitrage-free level. The estimates of the transition functions are shown on Figure 2 for selected stocks; they are plotted against the transition variable, q_{t-d} , (Panel A) and against time (Panel B). These figures visually support the nonlinear nature of the price spread series and the appropriateness of the ESTAR model, since, in general, observations seem to symmetrically lie above and below the parity. Again, we notice a relation between slow convergence and liquidity. For example, in Panel A, for a firm with a good daily volume like KOF, a previous day's deviation from parity of the order plus or minus 2%, the transition function attains smaller values (0.5), implying a

relatively slow mean reversion, whereas for a larger previous day's deviation around 4%, the transition function reaches the value of 1, the regime of full arbitrage, signaling a faster reversion. On the other hand, for a firm with a low daily volume TMM, a 30% spread makes the transition function equal to .5. In general, most of the transition functions in Panel A indicate that deviations lower than 5% trigger a full arbitrage regime.

V.III Half-lives and Convergence to Parity

While both estimated ESTAR models and transition functions shed lights on the nonlinear nature of the reversion of the price spread to parity, more insights into the adjustment mechanism of the models can be gained by estimating the average time it takes for a given shock to die out, also called the speed of convergence to parity. Following Taylor and Peel (2000) and Taylor et al. (2001), we estimated the half-lives²³ for shocks using the generalized impulse response function (GIRF)²⁴ introduced by Koop et al. (1996).

The half-life is defined in a non-linear framework as the number of periods taken by the impulse response function to fall below 0.5 d, or $GIRF < 0.5 d$, with $d =$

$\ln(1 + \frac{k}{100})$, where k represents the percentage of shocks.²⁵ Alternatively, to mitigate

²² They are constructed as in Eitrheim and Teräsvirta (1996).

²³ A half-life is defined as the number of periods it takes for shocks to the price spread to dissipate by half.

²⁴ In symbols, the generalized impulse response function is the difference between two conditional first moments:

$GIRF_x(j, s_t, \omega_{t-1}) = E[X_{t+j} | s_t, \omega_{t-1}] - E[X_{t+j} | \omega_{t-1}]$, $j=1,2,\dots,N$,

where $E[.]$ denotes the expectation operator, j is the forecasting horizon, s_t is the perturbation of the system at time t , $h_t = \omega_{t-1}$ represents the conditioning information set at time $t-1$ consisting of the history or initial conditions of the variable. $GIRF_x(j, s_t, \omega_{t-1})$ is computed using a dynamic stochastic simulation.

See also Peel and Venetis (2003a, 2003b) for a nice application of the generalized impulse response function to measure the half-lives of real exchange rates. See Koop et al. (1996) for technical details.

differences in GIRF due to the different variability of the underlying series, shocks can be set as $d = i\hat{\sigma}_\varepsilon$ where $\hat{\sigma}_\varepsilon$ denotes the residual standard deviations, $i=1, 3$, and 5 . We then estimated the half-lives for all price spread series for three sizes of shocks: $1\hat{\sigma}_\varepsilon$, $3\hat{\sigma}_\varepsilon$ and $5\hat{\sigma}_\varepsilon$. We also computed half-lives for a linear adjustment using the Augmented Dickey-Fuller (ADF) representation for the price spread:

$$(1-L)q_t = d_t + \alpha q_{t-1} + \sum_{j=1}^k \phi_j(1-L)q_{t-j} + \varepsilon_t,$$

where L denotes the lag operator, ε_t the error term, α , the persistence parameter, and d_t the deterministic component which can be a constant, m_0 , or a constant and a time trend, $\mu_0 + \beta t$; k denotes the autoregressive lag length and, ϕ_j the coefficient on k first differences. We use different lag specifications, for a maximum lag length of 5 and 20 business days.

In the second to fourth columns of Table 5, we report the estimated half-lives for all the firms, using the ESTAR model, for the three different sizes of shocks. In the last three columns, we also report the half-lives estimates for the standard AR linear adjustment model. All half-life estimates are expressed in business days. From the non-linear estimation, we observe faster adjustments for the majority of firms. The half-life estimates are similar across shock sizes, and a larger shock to the price spread triggers a faster reversion to parity. For the non-linear model, using one residual standard deviation as the shock, the average half-life is 3.1 business days, a reduction of more than half when compared to an average half-life for the linear model of 7.27 business days. That is, we observe for all firms a significant reduction in the half-life estimates when nonlinearities are incorporated into the arbitrage model. These averages, however, are influenced by a few large observations. Note that for 15 out of 21 firms, using the nonlinear model, it takes about 1 day for the ADR-underlying price spread to be reduced

by half. The half-life shrinks to less than 1 day if the shock size is three times the residual standard deviation. Some of the high half-life estimates correspond to companies that display very low volume (CDG, ICM, TMM). This finding is similar to the results reported in Rabinovitch et al. (2003), where low volume is associated with higher transaction costs, and in Roll, Schwartz, and Subrahmanyam (2004), where liquidity and lack of arbitrage opportunities are positively related.

V.IV Nonparametric Tests of Association between Liquidity and Convergence

To formally explore whether popular indicators of a firm's liquidity such as daily volume, market capitalization, and float are correlated with a firm's price spread parity convergence, a nonparametric Spearman rank correlation test is conducted.²⁶ The null hypothesis is that a firm's liquidity characteristics are not related to the speed of transition between regimes or the speed of convergence to parity against the alternative of them being associated.

Table 6.A shows the ranking of firms' liquidity indicators and Spearman rank correlation test results are reported in Table 6.B. The results indicate that the null

²⁶ The Spearman rank correlation coefficient is computed using the ranks as the paired measurements on the variables (x_i, x_j). The test statistic is therefore, assuming no ties in either the x or y observations²⁶, given by:

$$r_s = 1 - 6 \sum_{j=1}^N [R(x_{1,j}) - R(x_{2,j})]^2 / [N(N^2 - 1)], \quad |r_s| = 1$$

where N is the observation size (number of firms), $x_{i,j}$ ($i=1,2$) is the rank of the j^{th} observation of series x_i and $R(x_{i,j}) = 1$ for the smallest observed value of x_i . The test statistic r_s is asymptotically and normally distributed with mean zero and variance $1/N$.

hypothesis of no association can be rejected at the 1% level for all liquidity characteristics. Indeed, the average daily volume, market capitalization, and float are all positively correlated to the half-life and the speed of transition between regimes calculated using our non-linear estimators. Our non-linear estimates provide a better measure of liquidity than the standard linear estimates. The estimated correlations using the non-linear half-life estimates are substantially higher than the estimated correlations using the linear half-life estimates. For example, the correlations between market capitalization, average daily volume and float and the non-linear half-life estimates are .83, .58, and .66, respectively, while the correlation between the same liquidity indicators and the linear half-life estimates are .70, .40, and .51, respectively.

VI. Conclusions

In this paper we study the convergence between the prices of ADRs and the prices of the Mexican traded shares. We have a sample of 21 dually listed shares (listed in Mexico and in the U.S.), that are listed as level II or level III ADRs. Since both markets have similar trading hours in Mexico, standard arbitrage considerations should make persistent deviation from price parity rare. We estimate two different non-linear adjustment models, the LSTAR and ESTAR models, along with a standard linear model to estimate the convergence of the ADRs and the locally traded shares. From our estimation results, first, we reject the linear adjustment model; and, second, based on different tests, we select the ESTAR model. Using the ESTAR model, we are able to estimate the half-life of different shocks to price spreads. We find that price spreads tend to die out quickly in a nonlinear framework. For 15 out of 21 firms it takes about 1 day

for the ADR-underlying price spread to be reduced by half. Four firms, however, have high half-life estimates (seven days or more), and, in general, correspond to companies that display very low volume, and thus, arbitrage might be difficult to execute. The results of a nonparametric Spearman correlation tests confirm this finding as most firm's liquidity market indicators are positively correlated to the speed of convergence to parity. The sample average half-life is 3.1 business days. By allowing non-linear adjustments, the average half-life is reduced by more than 57%, when compared to the standard linear model.

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TABLE 1: DATA DESCRIPTION

ADR ISSUE	SYMBOL	EXCHANGE	RATIO	INDUSTRY	TYPE	EFF.DATE
AMERICA MOVIL SA DE CV- SERIES 'L'	AMX	NYSE	1:20	Wireless Comm.	Level II	8-Feb-01
CEMEX S.A. DE CV	CX	NYSE	1:5	Building Materials	Level II	1-Sep-99
COCA-COLA FEMSA 'L' SHARES	KOF	NYSE	1:10	Beverage	Level III	1-Sep-93
CORPORACION DURANGO	CDG	NYSE	1:2	Forest Products & Paper	Level III	1-Jul-94
DESC, S.A. DE C.V.	DES	NYSE	1:20	Auto Parts & Tires	Level III	20-Jul-94
EMPRESAS ICA, S.A. DE C.V.	ICA	NYSE	1:6	Heavy Construction	Level III	1-Apr-92
FOMENTO ECONOMICO MEXICANO, S.A. DE C.V.	FMX	NYSE	1:10	Beverage	Level II	11-Feb-04
GRUMA, S.A. DE C.V. - 'B' SHARES	GMK	NYSE	1:4	Food	Level II	6-Nov-98
GRUPO AEROPORTUARIO DEL SURESTE	ASR	NYSE	1:10	Gen. Industrial Svcs	Level III	28-Sep-00
GRUPO IMSA	IMY	NYSE	1:9	Industrial Diversified	Level III	10-Dec-96
GRUPO INDUSTRIAL MASECA S.A. DE C.V.	MSK	NYSE	1:15	Food	Level II	17-May -94
GRUPO IUSACELL	CEL	NYSE	1:5	Wireless Comm.	Level II	5-Aug-99
GRUPO RADIO CENTRO, S.A. DE C.V.	RC	NYSE	1:9	Broadcasting	Level III	9-Jul-93
GRUPO SIMEC 'B' SHARES	SIM	AMEX	1:1	Mining & Metals	Level III	1-Jun-93
GRUPO TELEVISA, S.A.	TV	NYSE	1:20	Broadcasting	Level III	16-Sep-02
GRUPO TMM	TMM	NYSE	1:1	Industrial Transport	Level III	17-Jun-92
INDUSTRIAS BACHOCO	IBA	NYSE	1:6	Food	Level III	26-Sep-97
INTERNACIONAL DE CERAMICA	ICM	NYSE	1:5	Building Materials	Level III	15-Dec-94
TELEFONOS DE MEXICO S.A. DE C.V.- SERIES 'L'	TMX	NYSE	1:20	Fixed Line Comm.	Level III	13-May -91
TV AZTECA, S.A. DE C.V.	TZA	NYSE	1:16	Broadcasting	Level III	1-Aug-97
VITRO, S.A. DE C.V.	VTO	NYSE	1:3	Industrial Diversified	Level III	19-Nov-91

TABLE 2: MARKET STATISTICS

SYMBOL	Volume ¹	MC	Float ²	Short Rate ²	Mean (QPar _t)	SD (QPar _t)	Max/Min	AR(1)	LB(5)
AMX	1,531,594	22.13B	386.70M	3.155	0.1154	1.0538	8.27/-9.46	0.391	207.31
CX	617,121	9.87B	139.84M	2.586	-0.4298	0.9613	5.44/-6.26	0.490	844.66
KOF	179,258	3.83B	27.08M	1.909	0.3720	1.8516	12.79/-9.82	0.414	615.21
CDG	23,641	55.10M	2.30M	5	16.575	58.1485	483.44/-52.00	0.988	116118.20
DES	50,501	665.14M	34.66M	2.667	-10.55	6.6358	14.01/-37.29	0.917	8659.15
ICA	266,505	615.47M	71.20M	16.645	-2.6896	2.4973	13.66/-17.53	0.456	1427.63
FMX	237,435	4.78B	66.26M	2.851	-0.1025	0.9973	7.56/-17.41	0.194	81.35
GMK	8,323	743.26M	21.68M	7.25	0.6519	3.9613	14.19/-81.40	0.479	758.81
ASR	130,500	587.40M	10.80M	0.407	0.8898	6.8541	86.48/-12.32	0.810	1172.07
IMY	19,590	1.13B	10.02M	9.5	0.5551	2.5380	28.56/-17.34	0.633	1842.60
MSK	33,240	397.98M	4.42M	N/A	0.5484	2.4037	14.38/-29.99	0.501	1539.18
CEL	150,975	116.56M	4.72M	18.2	-0.3614	4.4567	22.51/-29.53	0.667	1291.53
RC	37,263	98.50M	8.68M	2.042	2.7286	9.3029	64.47/-24.51	0.898	8233.55
SIM	11,000	298.47M	16.10M	1.978	1.0113	11.17	77.66/-59.88	0.874	8139.64
TV	692,304	5.98B	105.89M	2.884	0.0915	1.1807	16.28/-11.14	0.267	291.30
TMM	62,341	150.38M	6.30M	16.968	-0.6590	15.3707	101.31/-68.87	0.963	12768.54
IBA	18,586	471.38M	8.17M	1.294	0.1688	2.7932	17.90/-12.13	0.702	2054.45
ICM	6,945	107.68M	5.54M	N/A	3.5991	13.2577	64.95/-49.16	0.955	9441.14
TMX	2,520,634	19.09B	392.98M	8.095	0.6146	0.971	16.53/-10.86	0.305	730.84
TZA	440,805	1.49B	72.68M	7.42	-0.2084	1.4993	10.64/-10.26	0.309	259.07
VTO	164,141	304.66M	24.70M	4.636	0.3206	2.2561	14.81/-17.16	0.473	1819.54

Notes:

1. MC: Market Capitalization; Volume: average daily volume since inception, Float: number of freely traded shares in the hands of the public. Float is calculated as Shares Outstanding minus Shares Owned by Insiders, 5% Owners, and Rule 144 Shares. Mean(Par_t) is the mean of $QPar_t = 100 * (b S_t P_t(L) / P_t(A) - 1)$; SD(QPar_t) is the SD of QPar_t; Max/Min represents the maximum and the minimum of QPar_t; AR(1) is the AR(1) coefficient of QPar_t; and LB(5) is the Ljung-Box statistics with 5 lags for QPar_t.

2. As of May 18, 2004. N/A: Not available

TABLE 3: LM-Tests for Nonlinearity and LM-Tests for Model Selection

FIRM	Tests	Standard Tests					Heteroskedasticity-Robust Tests				
		AR vs STAR		ESTAR vs LSTAR			AR vs STAR		ESTAR vs LSTAR		
		TP	EJP	TP	EJP		TP	EJP	TP	EJP	
	d	NLM3	NLM4	NLM2	LM ^{LST}	LM ^{EST}	NLM3	NLM4	NLM2	LM ^{LST}	LM ^{EST}
Panel A TMX	1	0.000	0.000	0.000	0.000	0.000	0.387	0.390	0.042	0.016	0.002
	2	0.000	0.000	0.000	0.000	0.000	0.049	0.037	0.078	0.345	0.083
	3	0.000	0.000	0.000	0.000	0.000	0.008	0.601	0.138	0.098	0.092
	4	0.000	0.000	0.000	0.127	0.000	0.316	0.271	0.192	0.821	0.162
	5	0.000	0.000	0.002	0.009	0.043	0.263	0.426	0.220	0.404	0.430
Panel B											
AMX	4	0.000	0.000	0.001	0.001	0.000	0.021	0.037	0.002	0.035	0.017
CX	2	0.000	0.029	0.000	0.002	0.000	0.200	0.569	0.169	0.101	0.113
KOF	1	0.000	0.000	0.013	0.000	0.020	0.069	0.059	0.049	0.073	0.037
CDG	1	0.000	0.019	0.000	0.002	0.000	0.020	0.599	0.069	0.091	0.063
DES	1	0.000	0.000	0.010	0.000	0.020	0.054	0.070	0.041	0.093	0.057
ICA	1	0.000	0.000	0.015	0.000	0.000	0.000	0.000	0.576	0.519	0.530
FMX	1	0.000	0.000	0.004	0.032	0.000	0.072	0.042	0.051	0.051	0.014
GMK	1	0.000	0.000	0.027	0.000	0.644	0.030	0.051	0.094	0.045	0.002
ASR	1	0.000	0.000	0.000	0.000	0.000	0.112	0.066	0.032	0.071	0.033
IMY	1	0.000	0.000	0.000	0.000	0.000	0.202	0.034	0.053	0.054	0.032
MSK	2	0.000	0.007	0.000	0.026	0.003	0.046	0.270	0.037	0.074	0.055
CEL	4	0.000	0.019	0.001	0.053	0.000	0.057	0.031	0.095	0.099	0.023
RC	1	0.000	0.012	0.001	0.080	0.020	0.072	0.065	0.068	0.097	0.053
SIM	1	0.000	0.000	0.120	0.007	0.067	0.002	0.048	0.029	0.034	0.051
TV	1	0.000	0.000	0.000	0.001	0.000	0.045	0.055	0.032	0.107	0.092
TMM	1	0.000	0.000	0.000	0.000	0.000	0.004	0.027	0.076	0.088	0.078
IBA	2	0.000	0.001	0.010	0.001	0.016	0.032	0.019	0.164	0.026	0.014
ICM	1	0.000	0.000	0.000	0.000	0.000	0.151	0.046	0.105	0.062	0.043
TMX	2	0.000	0.000	0.000	0.000	0.000	0.049	0.037	0.078	0.345	0.083
TZA	5	0.000	0.000	0.004	0.000	0.070	0.013	0.018	0.057	0.081	0.043
VTO	2	0.000	0.000	0.268	0.025	0.082	0.080	0.067	0.063	0.081	0.052

Notes: This Table presents the p-values of the Lagrange Multiplier (LM) tests for AR linearity against STAR nonlinearity, denoted AR vs. STAR and LM-tests for choosing between the ESTAR and the LSTAR model, denoted ESTAR vs. STAR of the daily price differential between ADRs and their underlying shares. The tests are performed following two tests: the Teräsvirta (1994) test (TP) with the corresponding statistics NLM3, and NLM2; and the Escabano and Jorda (1999) test (EJP) with corresponding statistics NLM4, LM^{LST}, and LM^{EST}. The NLM3 and NLM2 statistics are based on the auxiliary regression model, equation (10) and the NLM4, LM^{LST}, and LM^{EST} statistics are based on equation (11). For each test, two versions of tests are estimated, the standard test and the heteroskedasticity-consistent test.

The first panel shows p-values for all possible choices of d, d={1,...,5} only for the firm TMX. The second panel also reports the selected p-values and the delay parameter, d, for all the other firms. For each test, the rejection of the null hypothesis, the selection of the delay parameter d, and the resultant model are based on the smallest p-value.

TABLE 4: Nonlinear Estimation Results for ESTAR model of Price Spread

$$\text{ESTAR(P): } q_t = \mu + \sum_{j=1}^p \alpha_j L^j(q_t - \mu) + \sum_{j=1}^p \hat{\alpha}_j L^j(q_t - \mu) \left[1 - \exp \left\{ -l (q_{t-d} - \mu)^2 / \hat{\sigma}_{z_t}^2 \right\} \right] + e_t$$

FIRM	p, d	μ	l	Y ₁₁	Y ₁₂	Y ₂₁	Y ₂₂	S	LR(k)	NL _{Max} d={3..6}	LMNA AR(1-6)
AMX	2, 4	-	2.745 (0.019)	0.842 (0.106)	0.155 (0.082)	-0.837 (0.547)	-0.137 (0.544)	0.453	LR(4) [0.561]	[0.498]	[0.336]
CX	2, 2	-	2.864 (0.008)	0.494 (0.066)	0.259 (0.046)	-0.47 (0.108)	-0.258 (0.114)	0.061	LR(4) [0.754]	[0.502]	[0.452]
KOF	1, 1	0.045 (0.001)	1.575 (0.025)	0.643 (0.049)	-	-0.612 (0.077)	-	0.036	LR(1) [0.224]	[0.471]	[0.357]
CDG	1, 1	-	1.981 (0.528)	0.993 (0.009)	-	-0.996 (0.048)	-	0.041	LR(3) [0.582]	[0.211]	[0.405]
DES	2, 1	0.163 (0.003)	2.971 (0.068)	0.921 (0.050)	0.164 (0.048)	-0.928 (0.062)	0.141 (0.066)	0.028	LR(3) [0.672]	[0.404]	[0.471]
ICA	2, 1	-0.221 (0.012)	0.992 (0.082)	-0.765 (0.113)	-0.213 (0.108)	0.548 (0.082)	0.098 (0.074)	0.039	LR(3) [0.423]	[0.397]	[0.545]
FMX	2, 1	-	1.793 (0.014)	0.865 (0.011)	0.147 (0.075)	-0.859 (0.070)	-0.161 (0.094)	0.027	LR(4) [0.522]	[0.438]	[0.443]
GMK	1,1	-	0.361 (0.039)	0.839 (0.234)	-	-0.841 (0.043)	-	0.042	LR(2) [0.252]	[0.399]	[0.562]
ASR	1, 1	-	1.277 (0.027)	0.806 (0.122)	-	-0.809 (0.439)	-	0.021	LR(2) [0.252]	[0.305]	[0.668]
IMY	2, 2	-	0.642 (0.030)	0.526 (0.121)	0.164 (0.113)	-0.535 (0.109)	0.160 (0.114)	0.048	LR(3) [0.352]	[0.574]	[0.218]

Notes:

P and d denote the autoregressive order and the number of period for the delay parameter respectively. Y₁₁, Y₁₂, Y₂₁ and Y₂₂ represent the estimated autocorrelation parameters, l the estimated speed of transition, μ the estimated mean, and S the residual standard errors of models.

Figures reported in the squared brackets are the (p-values). ARCH tests conducted on the residuals of the estimated models indicated the presence of heteroscedasticity. Therefore numbers in parentheses denote heteroscedastic-consistent standard errors of estimates computed using Woodridge (1991). LMNA AR(1-6) is a LM-test statistics for testing the null hypothesis of no serial correlation in the residuals of order 1, up to 6 developed as in Eitrheim and Teräsvirta (1996). NL_{Max} denotes the maximum LM - test statistic of no additive nonlinearity with the delay length in the range from 3 to 6; they are constructed as in Eitrheim and Teräsvirta (1996). LMNA and NL_{Max} allow the assessment of models adequacy. LR(k) denotes a likelihood ratio test statistic for k parameters restrictions implicit to the estimated equation against the unrestricted ESTAR model. For instance, LR(4) indicates the test statistic for imposing four restrictions $\mu = 0$, $\Psi_{11} + \Psi_{12} = 1$, $\Psi_{21} = -\Psi_{11}$, and $\Psi_{22} = -\Psi_{12}$, on the models and re-estimate them if these restrictions are not rejected by the data.

TABLE 4 : (Continued) Nonlinear Estimation Results for ESTAR model of Price Spread

FIRM	p, d	μ	l	Y_{11}	Y_{12}	Y_{21}	Y_{22}	S	LR(k)	NL_{Max} d={3..6}	LMNA AR(1-6)
MSK	2, 2	0.025 (0.003)	0.582 (0.053)	0.897 (0.111)	0.175 (0.077)	-0.876 (0.094)	-0.133 (0.083)	0.030	LR(3) [0.571]	[0.327]	[0.525]
CEL	2, 4	0.006 (0.026)	0.496 (0.038)	0.611 (0.071)	0.305 (0.066)	-0.609 (0.123)	-0.095 (0.132)	0.041	LR(3) [0.471]	[0.318]	[0.280]
RC	1, 2	0.211 (0.012)	0.835 (0.104)	0.876 (0.031)	-	-0.950 (0.043)	-	0.038	LR(1) [0.197]	[0.390]	[0.572]
SIM	2, 1	0.348 (0.010)	0.514 (0.099)	0.911 (0.084)	0.234 (0.065)	-1.09 (0.183)	-0.239 (0.173)	0.052	LR(3) [0.458]	[0.795]	[0.489]
TV	2, 1	-	2.289 (0.030)	0.823 (0.158)	0.081 (0.139)	-0.824 (0.101)	-0.078 (0.073)	0.032	LR(4) [0.285]	[0.321]	[0.254]
TMM	1, 1	-0.294 (0.711)	0.315 (0.737)	0.973 (0.019)	-	-0.973 (0.019)	-	0.042	LR(1) [0.628]	[0.244]	[0.145]
IBA	2, 2	-	1.550 (0.021)	0.638 (0.038)	0.296 (0.031)	-0.641 (0.274)	-0.278 (0.276)	0.019	LR(4) [0.356]	[0.275]	[0.323]
ICM	1, 1	-0.053 (0.014)	0.317 (0.094)	0.975 (0.010)	-	-0.846 (0.039)	-	0.035	LR(1) [0.334]	[0.399]	[0.379]
TMX	2, 2	0.026 0.000	2.853 (0.018)	0.718 (0.065)	0.261 (0.057)	-0.709 (0.456)	-0.131 (0.173)	0.010	LR(3) [0.573]	[0.589]	[0.258]
TZA	2, 5	0.028 (0.047)	1.387 (0.013)	0.965 (0.042)	0.023 (0.042)	-0.978 (0.149)	0.127 (0.135)	0.031	LR(3) [0.628]	[0.535]	[0.332]
VTO	2, 2	-0.037 (0.005)	0.984 (0.093)	0.771 (0.092)	0.156 (0.058)	-0.725 (0.065)	-0.134 (0.048)	0.019	LR(4) [0.425]	[0.419]	[0.425]

TABLE 5: Speed of Convergence: Half-lives

Nonlinear Adjustment (ESTAR) ^a				Linear Adjustment (AR) ^b		
FIRM	$1\hat{\sigma}_\varepsilon$	$3\hat{\sigma}_\varepsilon$	$5\hat{\sigma}_\varepsilon$	$q_t(5)$	$q_t(20)$	$q_t^\mu(20)$
AMX	0.643	0.588	0.507	0.850	0.850	0.845
CX	0.516	0.544	0.514	2.318	4.253	1.854
KOF	0.701	0.634	0.612	0.839	0.932	0.932
CDG*	10.895	10.759	9.661	50.027	50.027	45.045
DES	2.356	2.267	1.084	12.313	14.046	10.389
ICA*	12.908	12.772	11.674	15.685	78.779	47.324
FMX	0.542	0.497	0.494	0.505	0.597	0.484
GMK	1.079	0.968	0.555	0.732	1.240	0.637
ASR	0.945	0.892	0.712	2.287	2.797	2.740
IMY	0.986	0.866	0.793	2.057	2.235	2.231
MSK	1.042	1.045	0.947	1.691	2.513	1.860
CEL	2.094	2.195	2.183	2.601	4.259	4.946
RC	3.616	3.527	2.344	7.014	7.014	7.012
SIM	1.893	1.846	1.008	7.363	8.654	8.508
TV	0.694	0.664	0.666	0.704	0.704	0.674
TMM*	12.575	12.439	11.341	32.550	28.247	26.363
IBA	1.995	1.764	1.103	2.344	2.344	2.079
ICM*	7.116	7.027	5.844	7.014	23.981	23.317
TMX	0.551	0.458	0.475	1.040	2.039	1.643
TZA	0.606	0.603	0.603	0.792	1.138	1.063
VTO	1.164	0.976	0.832	1.799	3.165	2.375

Notes: All figures are in (business) days. A half-life is defined as the number of periods it takes for shocks to pricing error to dissipate by a half. In a non-linear framework, it is such that the impulse response function is less than unity or $G_h(d, ?_{t-1}) < 0.5$

a. Half-lives for shocks $\hat{\delta} = i\hat{\sigma}_\varepsilon$ ($i=1,3,5$) where $\hat{\sigma}_\varepsilon$ denotes the residual standard deviation

b. Half-lives computed in a linear framework, using the Augmented Dickey-Fuller (ADF) representation, that regresses the first difference of the price spread q_t on a deterministic component, its lagged level, and k lagged first differences:

$$(1-L)q_t = \alpha + a_1 q_{t-1} + \sum_{j=1}^k f_j(1-L)q_{t-j} + e_t$$

where L denotes the lag operator, e_t the error term, a_1 the persistence parameter, and α the deterministic component which can be a constant, m_0 , or a constant and a time trend, $m_0 + \beta t$, and k denotes the autoregressive lag length. The maximum lag length in the ADF specification is set equal to 5 and 20 business days. The lag truncation is selected using a general-to-specific methods. In q_t^μ , μ indicates that a time trend is incorporated in the determinist part of the pricing error process. * denotes firms with unusually higher half-life.

Table 6.A: Ranks of Firms Market Characteristics

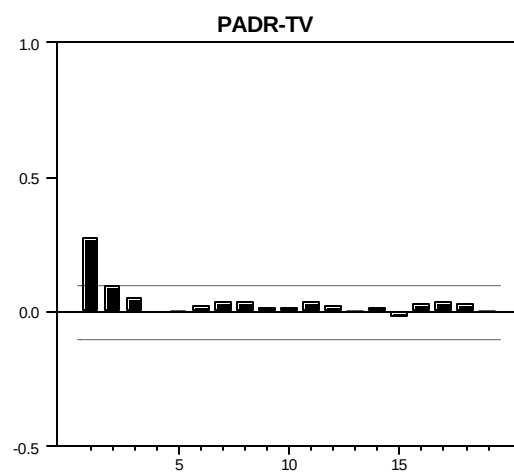
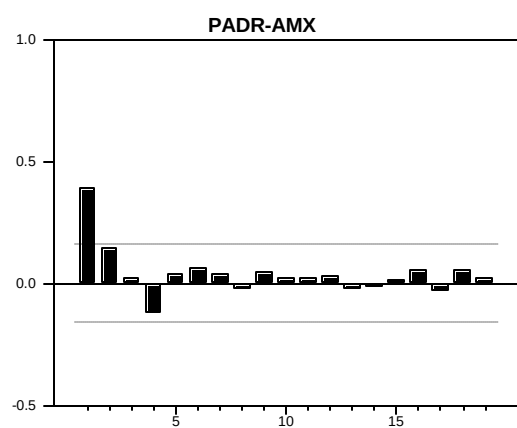
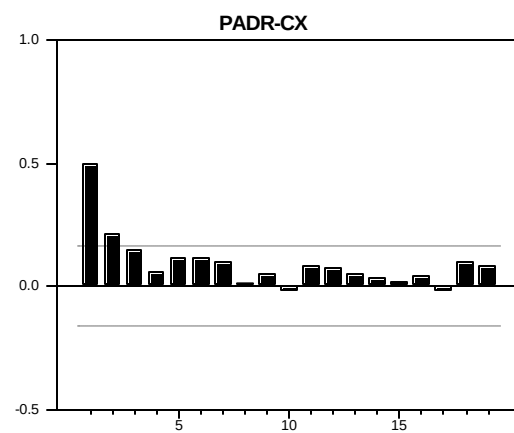
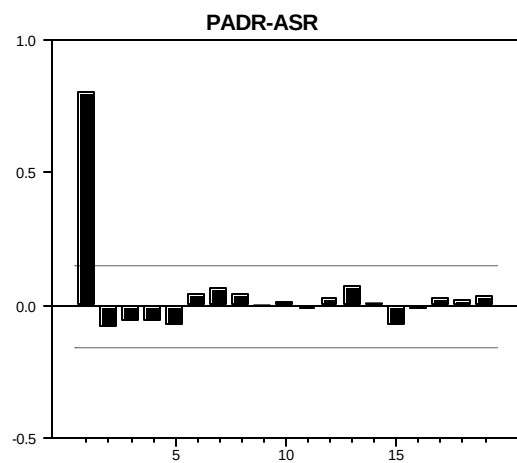
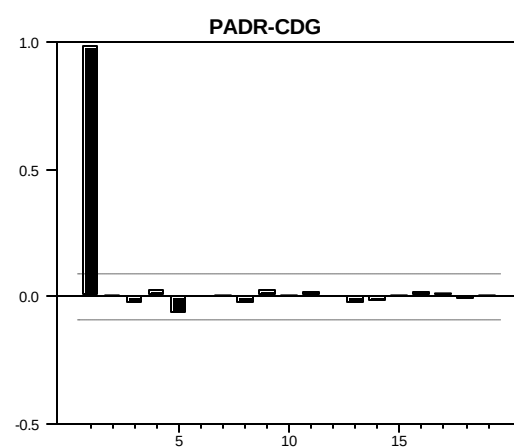
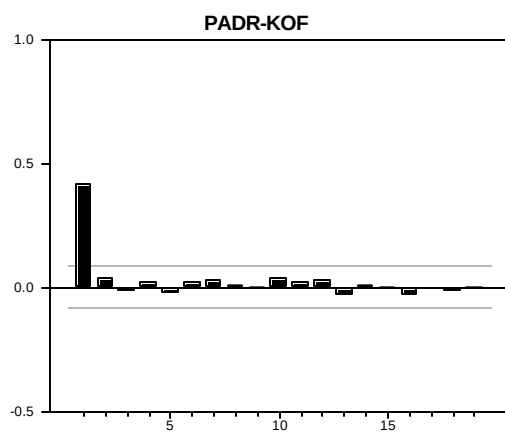
FIRM	Speed of transition	Price spread Nonlinear Half-life	Price spread Linear Half-life	Average Daily Volume	Market Capitalization	Float
AMX	4	5	6	2	1	2
CX	2	1	12	4	3	3
KOF	8	7	5	8	6	9
CDG	6	19	21	16	21	21
DES	1	16	18	13	10	8
ICA	12	21	19	6	11	6
FMX	7	2	1	7	5	7
GMK	19	11	3	20	9	11
ASR	11	8	11	11	12	13
IMY	15	9	10	17	8	14
MSK	16	10	8	15	14	20
CEL	18	15	14	10	18	19
RC	14	17	15	14	20	15
SIM	17	13	17	19	16	12
TV	5	6	2	3	4	4
TMM	21	20	20	12	17	17
IBA	9	14	13	18	13	16
ICM	20	18	16	21	19	18
TMX	3	3	7	1	2	1
TZA	10	4	4	5	7	5
VTO	13	12	9	9	15	10

Table 6.B: Nonparametric Tests of Association Between Firm Market Characteristics and Convergence to Parity: Spearman Rank Correlation Coefficient (r_s)

	Average Daily Volume	Price spread Nonlinear Half-life	Price spread Linear Half-life	Market Capitalization	Float
Speed of Transition	0.627	0.513	0.223	0.633	0.651
Average Daily Volume		0.579	0.404	0.513	0.777
Price spread Nonlinear Half-life			0.810	0.826	0.655
Price spread Linear Half-life				0.702	0.505
Market Capitalization					0.852

Notes: * denotes significance at the 5% level. The Spearman rank statistics indicate that volume, market capitalization, and float are positively and strongly correlated with the price spread half live and the speed of transition. This implies the higher the average daily volume, the faster an arbitrage can be executed. This observation remains true for market capitalization and float. Critical values of Spearman's Rank correlation coefficient for n=21 are: 0.368 (5%); 0.438(2.5%); and 0.521(1%).

Figure 1: Partial Autocorrelation Function for q_t for Selected Firms

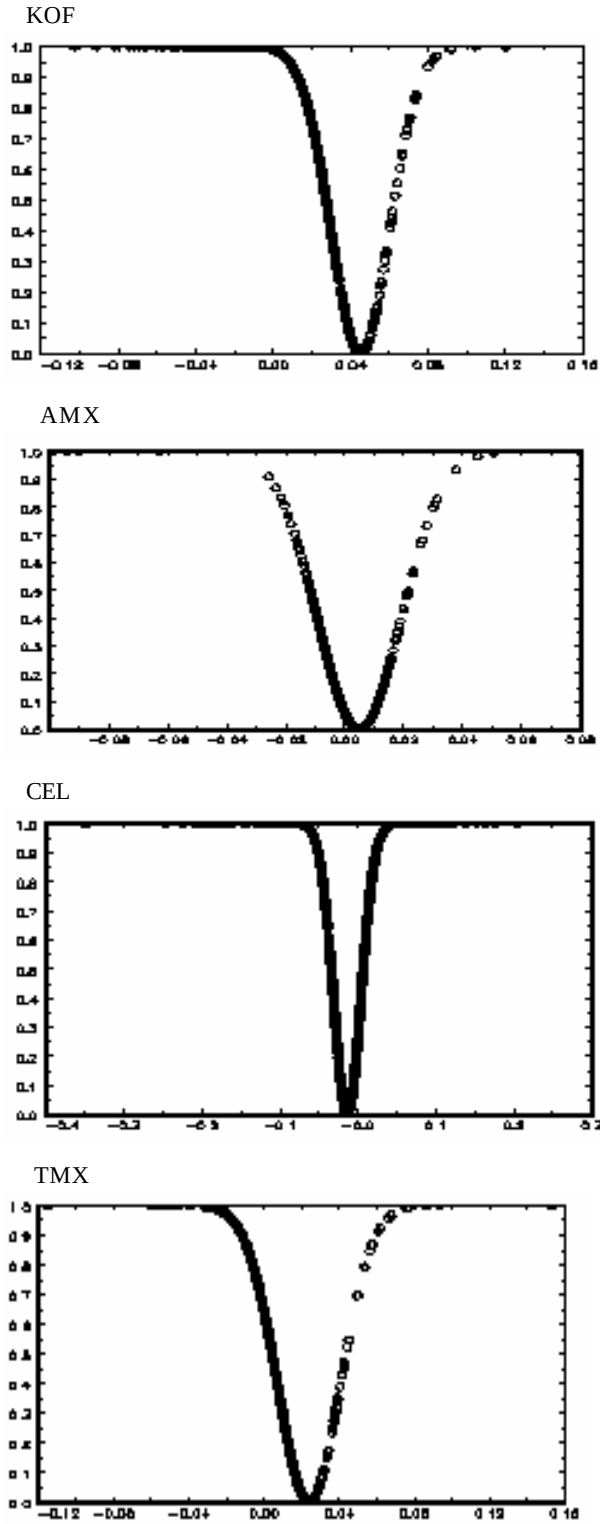


Notes: The band on the PACF represents the 95% confidence interval.

Figure 2: Estimated Transition Function for Selected Firms

Panel A: Transition Function vs. Transition Variable

$$F(Z_{t-d}; l, \mu) \text{ v.s. } Z_{t-d}$$



Panel B: Transition Function vs. Time

$$F(Z_{t-d}; l, \mu) \text{ v.s. } t$$

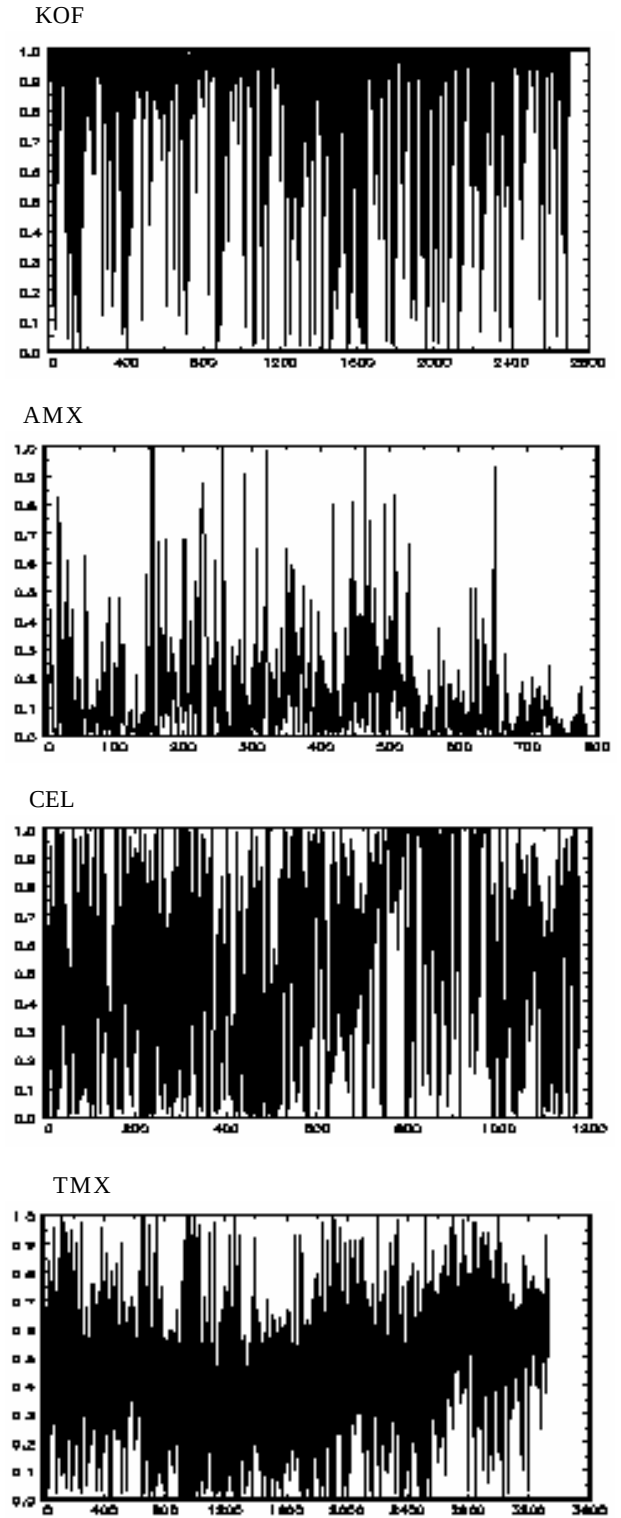
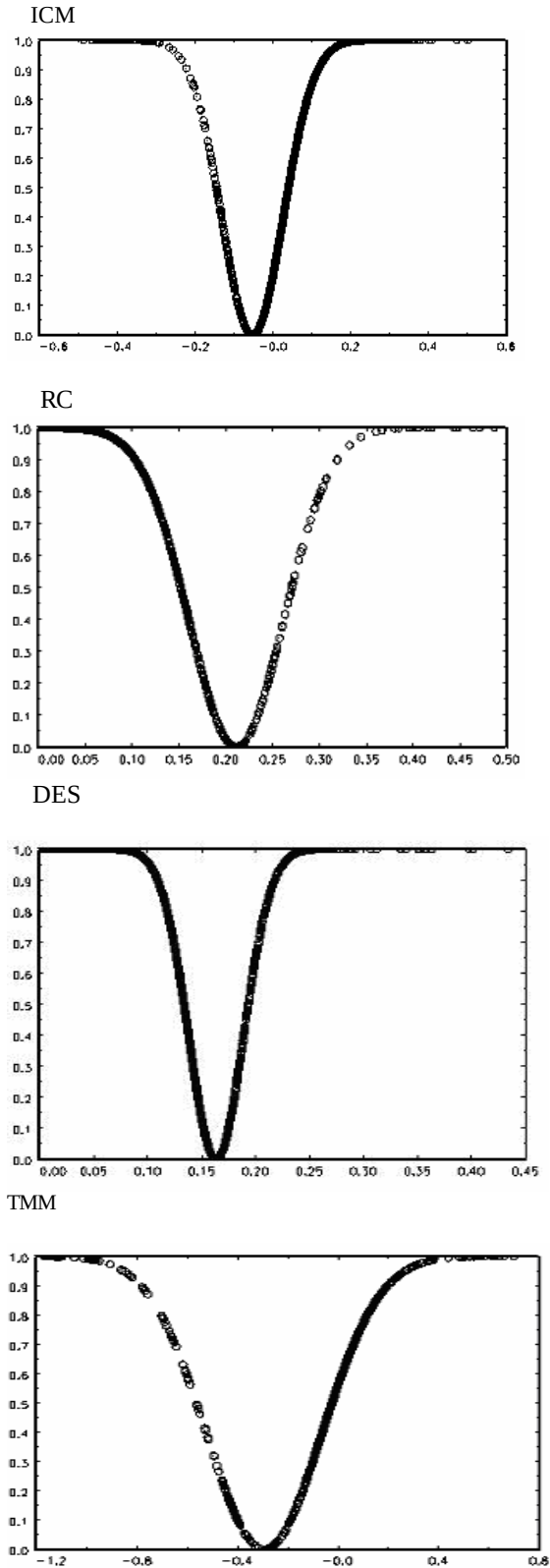


Figure 2: Estimated Transition Function for Selected Firms (continued)

Panel A: Transition Function vs. Transition Variable

$$F(Z_{t-d}; l, \mu) \text{ v.s. } z_{t-d}$$



Panel B: Transition Function vs. Time

$$F(Z_{t-d}; l, \mu) \text{ v.s. } t$$

